

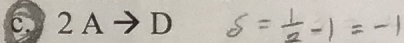
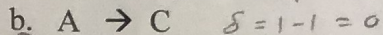
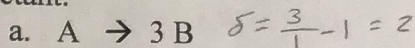
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CHE 321; Spring - 2020; Quiz - 3

100 Points

90

1. (25 points) Three gas phase reactions have the same reaction rate dependence to the concentration of the reactant.



$$V = V_0 (1 + \epsilon x) \left( \frac{T}{T_0} \right)^{\gamma} \left( \frac{P_0}{P} \right)^{\delta}$$

isothermal  
isobaric

$$\epsilon = \sum \nu_i \delta_i$$

$$\delta = \frac{\nu}{a} - 1$$

Which of these reactions will exhibit the highest conversion in a PFR? Explain.

c has the highest conversion because  $V < V_0$  since  $\delta = -1$  which increases reaction time and because volume decreases, it increases reactant concentration, which both increase conversion.

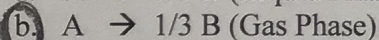
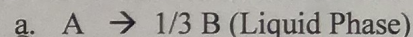
2. (25 points) Why does the conversion for a gas-phase reaction in a PBR depend on the pressure?

$$C_A = \frac{F_A}{V}$$

$$V = V_0 \frac{F_T}{F_{T0}} \left( \frac{P_0}{P} \right) \left( \frac{T_0}{T} \right)$$

Yes because concentration depends on pressure.

3. (25 points) Two reactions (one gas phase and one liquid phase) have the same reaction rate dependence to the concentration of the reactant.



Which of these reactions will exhibit the highest conversion in a PFR? Explain.

$V < V_0$  for gas, so the reactant concentration increases and the reaction time which both increase conversion.

$V \downarrow$  Concentration  $\uparrow$   $\tau \uparrow$   
reactant

4. (25 points) A reaction rate equation is given by  $r_A' = k' P_A$ . What are the units of  $k'$ ?

$$r_A' = \frac{\text{moles}}{(\text{time})(\text{mass})}$$

$$P_A = \text{Pressure}$$

$$k' = \frac{r_A'}{P_A} = \frac{\text{moles}}{(\text{time})(\text{mass})(\text{Pressure})}$$